An Auditable Public-Data U.S. Equity Research Pipeline: Point-in-Time Factors, Asset Pricing, Portfolio Construction, and Walk-Forward Machine Learning

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Revision IMFP-P4-2026-11-01 | 1 November 2026

Reconciled working-paper revision for final author review.

Abstract

This paper studies an auditable public-data-oriented U.S. equity research workflow linking security screening, SEC-derived characteristics, internally constructed factor portfolios, asset-pricing models, constrained allocations, and walk-forward prediction. The frozen sample screens 822 securities, accepts 487, and contains 715,447 market observations and 37,273 fundamental observations. Of 48 configured factor names, 47 have nonmissing raw values; these are neither independent signals nor uniformly available observations. Corrected holding endpoints produce 11,352 quantile-portfolio observations. On matched per-asset samples, mean adjusted R-squared ranges from 0.760818 for CAPM to 0.817658 for FF5, with negative mean cycle intercepts and no intercept surviving supplementary Holm adjustment across 615 tests. Eight long-only allocation methods are evaluated over 31 complete monthly trading cycles. CVaR has the highest observed cumulative net return, 73.68%, but descriptive rankings do not establish durable superiority. The corrected two-characteristic ML panel has 27,632 rows, 89 common monthly dates, and 15 dependent test folds. Exploratory mean-IC intervals are nominally positive for the linear families, cross zero for random forest, and are negative for XGBoost. Stable incremental after-cost economic value has not been established: the implemented economic check holds selected equities for one subsequent session, not a continuous 21-session strategy. The contribution is integrated, traceable empirical evaluation with explicit source-vintage, timing, dependence, and implementation limits.

Keywords: empirical asset pricing; point-in-time controls; factor proxies; portfolio construction; transaction costs; walk-forward validation; reproducibility. JEL classifications: C52, C55, G11, G12, G17.

I Introduction

Empirical finance often begins after difficult data decisions have been made. Security identity, delisting coverage, filing availability, corporate-action treatment, and provider transitions determine which observations a model sees. Portfolio and machine-learning layers add cost, tuning, temporal-split, and interpretation choices. This study brings those choices into one versioned workflow and asks what conclusions remain defensible after the evidence is bounded explicitly.

The project uses free and public sources rather than CRSP or Compustat. That improves access to tools and documentation but does not make the selected universe representative or the original source vintage publicly recoverable. Identity and lifecycle checks can identify problems among supplied securities; they cannot restore firms absent from the provider snapshot. Accounting period ends are distinct from filing acceptance and local capture times. The research problem is statistical and evidentiary.

The original questions are whether characteristics can be constructed on the selected panel, whether multifactor models improve explanatory fit, how constrained allocations differ under fixed costs, whether higher costs reduce cumulative outcomes, and whether two-characteristic models add stable out-of-sample information and economic value. The contribution is a connected empirical workflow, not a new asset-pricing theorem. Non-estimability, negative forecasts, and incomplete hypothesis decisions remain substantive findings.

This revision preserves the historical sequence from literature and source controls to hypotheses, factor construction, pricing, portfolios, ML, robustness, assessment, discussion, limitations, and conclusion. Appendices retain the definition catalog, statistics, methods, configurations, lineage, hypothesis evidence, software assurance, and equations. Historical manuscripts and publications remain archived; corrected tables supersede their empirical interpretation where the underlying experiment changed.

II Literature review

Markowitz [1] formalized mean-variance choice, and Sharpe [2] related expected returns to systematic risk. Fama and French [3] extended empirical modeling with size and value; Carhart [4] added momentum; Fama and French [5] introduced profitability and investment; Hou, Xue, and Zhang [6] developed an investment-based q-factor account. These motivate the study specifications. Their names do not establish that this project replicates official factor returns or identifies independent causal risks.

Expected-return and covariance estimation affect optimized portfolios. DeMiguel, Garlappi, and Uppal [7] show why simple diversification can be difficult to beat out of sample. Equal-risk-contribution portfolios [8], hierarchical risk parity [9], and the conditional-value-at-risk formulation [10] motivate comparisons across risk budgeting, clustering, and tail-loss objectives. Observed differences in this short experiment are not a general ranking of these methods.

White [11] and Newey and West [12] motivate heteroskedasticity-robust and heteroskedasticity-and-autocorrelation-consistent inference. A large factor registry creates multiplicity concerns [13]. Supplementary Holm adjustments in this audit qualify model intercept tests; they do not substitute for the missing original factor-level H1 decision procedure.

Ridge [14], Elastic Net [15], random forests [16], and XGBoost [17] supply regularized and nonlinear methods. Gu, Kelly, and Xiu [18] motivate ML in empirical asset pricing. Backtest-overfitting concerns [19] and temporal-validation methods [20] motivate chronological training, validation, purging, and embargo. SHAP [21] explains fitted predictions; attribution does not identify a causal effect or stable investment rule.

III Data sources and source limitations

Table I Data sources and evidence boundaries

Source	Role	Evidence boundary
HF Data Library [22,27]	Historical stock bars and candidate universe	Frozen historical vintage; current release differs
SEC EDGAR [23]	Submissions, Company Facts and identity	Acceptance and capture chronology are distinct
Alpha Vantage [24]	Listing and lifecycle evidence	Supplied symbols are not an exhaustive security master
FRED DGS3MO [25]	Treasury-yield reference for cash	Yield requires conversion to period returns
French Data Library [26]	Factor concepts and conventions	Project uses internally constructed proxies
SPY	Market and benchmark proxy	Complete dividend treatment unverified

HF Data Library supplied the candidate set and histories used locally. The historical evidence records a PiTrading/IEX source transition around March 2022. Provider documentation checked for this revision describes a restructured IEX-only release and a security list that is not reconstructed point in time [22,27]. That current description is not substituted for the archived dataset. Current availability and adjustment documentation do not certify every historical frozen quote.

The return calculations use frozen provider-vintage adjusted quotes. Complete dividend-inclusive total-return treatment is not independently established. This qualification applies to securities, constructed portfolios, and the SPY proxy. Current documentation characterizes IEX as approximately 2-3% of consolidated volume [27]; this is provider context, not an estimated historical coverage ratio. Dollar volume, share turnover, and liquidity characteristics require caution across the splice.

SEC APIs supply submissions and XBRL facts [23]. The design retains accession, taxonomy, concept, unit, and recorded availability. Period end cannot authorize earlier use. Amendments, local capture, issuer-to-share-class allocation, and receipt times remain incompletely verified across the full panel. Point-in-time-oriented controls are present; exhaustive historical certification is not claimed.

Alpha Vantage lifecycle evidence can extend before the empirical market panel [24]; lifecycle coverage is not return coverage. DGS3MO is a published annual yield reference, not a daily return [25]. Frozen normalized cash returns are compounded over exact portfolio holding intervals. French library conventions inform terminology; official downloaded factors do not replace the internal proxies.

IV Sample construction and point in time controls

Table II Universe screening outcomes

Disposition	Securities	Share
Screened	822	100.00%
Accepted	487	59.25%
Rejected	335	40.75%

Screening uses attribution, lifecycle, issuer/listing identity, shares, split basis, terminal events, asset eligibility, venue, and fundamental evidence. The fractions describe supplied candidates, not an exhaustive historical market. A source-selected panel can retain survivorship and selection effects even when each supplied candidate is screened.

Effective-dated listing identity is distinct from issuer identity. Ticker changes, share classes, venues, and succession require explicit mappings. Incomplete mapping and share evidence remain limitations. Unresolved exits are exclusions; missing delisting payouts are not replaced with invented or zero returns.

The market panel spans 14 February 2018-4 August 2026. Recorded fundamental availability spans 23 February 2019-15 August 2026. Later-available facts cannot justify earlier decisions. Final joins pass recorded-day availability gates, but these gates do not establish same-close executable information or exhaustive SEC capture chronology.

Table III Annual daily security breadth

Year	Min	Median	Mean	Max
2018	1	4	3.7	4
2019	4	14	17.8	46
2020	46	333	295.5	357
2021	357	398	393.3	404
2022	404	428	425.2	431
2023	431	463	457.6	466
2024	466	486	483.3	487
2025	487	487	487.0	487
2026	486	487	487.0	487

Note. The 2026 row ends on 4 August. Counts are distinct security IDs per observed date.

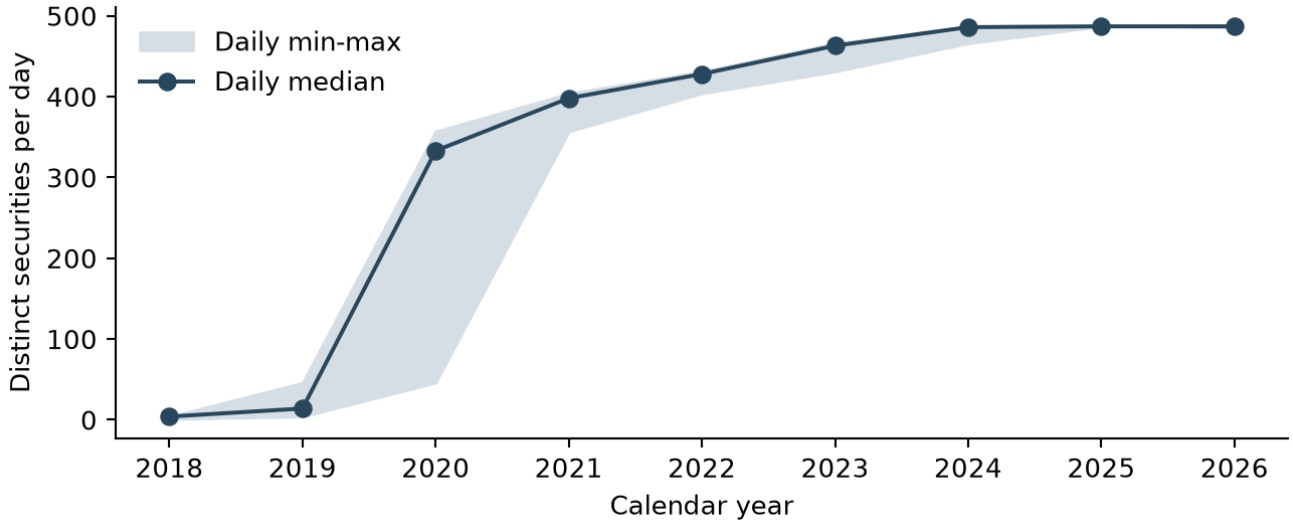


Figure 1. Annual daily security breadth in the frozen panel. Median and range summarize Table III; early cross-sections are very small.

Only one to four securities appear per observed day in 2018; the median in 2019 is 14. Breadth expands in 2020 and reaches 487 in 2024. This is an evidence-availability pattern, not proof of economically comparable cross-sections throughout. Minimum guards do not make singleton quantiles diversified.

Table IV Descriptive statistics of frozen inputs

Variable	N	Mean	SD	Median
security return	715,447	0.000587	0.026123	0.000575
market capitalization	706,285	5.3694e+10	1.8034e+11	2.1601e+10
daily dollar volume	715,447	6.2430e+07	3.0616e+08	1.2678e+07
book to market	585,078	0.395473	0.440632	0.292552
momentum 12 1m	593,210	0.124992	0.444258	0.069121
rolling volatility	706,194	0.389935	0.209722	0.329544

Note. Security return is daily decimal; capitalization is USD; daily dollar volume is price times volume in USD per day, not the rolling average-dollar-volume characteristic. Book-to-market is a ratio, momentum is cumulative decimal, and volatility is annualized. Appendix C reports quantiles.

A limited direct tabulation of authenticated frozen inputs reproduces the original descriptive values. It does not rerun factor construction or certify vendor adjustments. Market capitalization is frozen price times recorded shares; comparability of their bases remains conditional on source evidence.

V Research design and hypotheses

Source evidence feeds normalized inputs, characteristics, internal portfolios, regressions, allocations, and forecasts. Later computations consume identified parents without overwriting them. This dependency order is a software control; it does not establish prospective preregistration or historical approval dates.

Table V Hypotheses as worded in the supplied SSRN manuscript

ID	Original proposition
H1	At least one factor spread survives the complete pre-specified cost and multiplicity rule
H2	Multifactor models improve explanatory fit relative to CAPM and that improvement extends to stable out-of-sample economic evidence
H3	Estimated relations are stable through time under the original rule
H4	Higher transaction-cost assumptions reduce cumulative outcomes
H5	Constrained methods alter risk-return trade-offs and any apparent superiority is stable out of sample
H6	ML adds stable OOS predictive and after-cost value
H7	Explanation rankings are stable across methods

Note. Historical wording is preserved; original approval/decision evidence is not authenticated for H1/H3. The older 30-page paper has shorter H2/H5 wording. Appendix I records version differences; no retrospective rule is asserted.



Figure 2. Analytical dependency structure, not prospective preregistration. The portfolio-to-ML arrow denotes fee context only; ML also uses raw factor scores and market labels.

Missing hypothesis evidence is not replaced with favorable descriptive tests. H1 needs complete original factor-level costs and multiplicity evidence. H3 needs an authenticated direction and original stability rule. Rolling coefficients exist, but the accepted audit found a wording/direction discrepancy and no complete approved decision record. Both remain inconclusive.

VI Factor construction and results

The seven registry families are market, size, value, momentum, profitability, investment/leverage, and risk/liquidity. Forty-seven of 48 names have nonmissing raw values; names can map to related or duplicated exposures. Equity issuance remains non-estimable because issuance and equity inputs cannot be reconciled without unsupported imputation.

Table VI Factor construction scale and holding coverage

Measure	Count
Configured names	48
Definitions with nonmissing raw values	47
Physical long-form rows	34,341,456
Nonmissing raw values	28,550,589
Nonmissing factor scores	24,255,559
Quantile characteristics	41
Dependent portfolio IDs	123
Historical portfolio rows	11,545
Corrected complete-cycle rows	11,352

Note. Physical rows include missing values. Definitions, scores, characteristics and unique exposures have different denominators.

Preprocessing retains the released winsorization and robust normalization configuration, with a minimum eligible cross-section of three. Value-weighted terciles use eligible scores and positive normalized capitalization weights. Appendix A retains the definition catalog. Minimum counts are implementation guards, not representative diversification.

Accepted method full-xnys-delayed-close-endpoint-v2 uses month-end signals with UTC-day cutoff, modeled next-XNYS-session close entry, and exit at the following monthly execution close. Membership and weights are fixed before future-return inspection. Unavailable genuine endpoints reject a whole quantile; there is no survivor re-ranking, stale mark, zero fill, or invented terminal payout. These are complete modeled trading cycles, not verified fills.

Of 265 flagged historical groups, 195 have valid interior-gap return chains and 70 lack intended old endpoints. Intermediate gaps do not automatically omit buy-and-hold PnL. Seventy split-end consolidations and 123 terminal partial-cycle exclusions yield 11,352 rows. The changed delayed-close experiment does not recover unavailable old observations. The holding audit still contains 956 singleton quantile observations. Construction scale does not supply the missing original H1 cost/FDR decision.

VII Asset pricing models and results

Each model regresses portfolio excess returns on aligned internal factor proxies. CAPM uses market excess; FF3 adds SMB/HML; Carhart 4 adds MOM; FF5 adds RMW/CMA. The q-style specification uses size from log

capitalization, investment from asset growth, and a historical ROE proxy from earnings yield. This is study specific, not an exact Hou-Xue-Zhang replication or an accounting ROE factor [6].

OLS and three-lag Newey-West HAC intercept inference are retained. The corrected primary comparison has 123 assets and 9,441 asset-cycle observations per model, with 67-77 matched observations per asset and exits from 1 April 2020 to 3 August 2026. Each asset uses identical dates across models. The model-specific supplementary publication is separate and does not supply primary-table cells.

Table VII Corrected matched sample pricing comparison

Model	Mean adj R^2	Mean alpha	Nominal p < .05
CAPM	0.760818	-0.001658	8/123
FF3	0.802438	-0.001967	12/123
Carhart 4	0.811021	-0.001879	14/123
FF5	0.817658	-0.002008	8/123
q-style	0.814361	-0.002227	9/123

Note. Alpha is decimal per full monthly trading cycle, not annualized. Nominal counts use HAC p < .05 without multiplicity correction; all five models use the same matched per-asset dates.

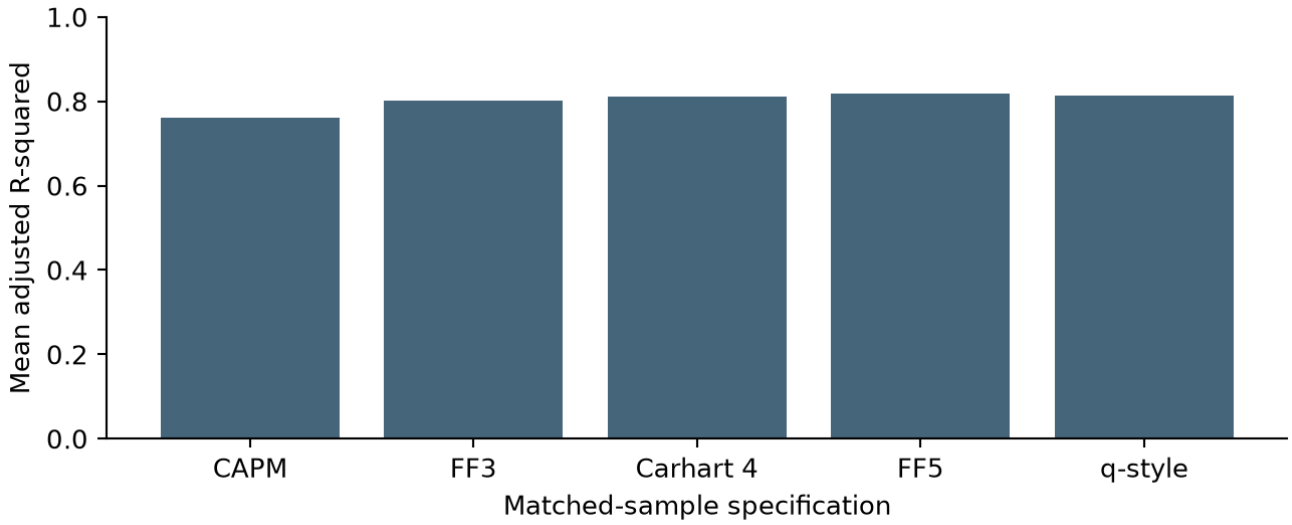


Figure 3. Corrected matched-sample mean adjusted R-squared. Fit is conditional on internal proxies and dependent test portfolios.

The corrected FF5-CAPM mean adjusted-R-squared difference is 0.056840; the historical 0.238 belongs to a superseded interval experiment. High fit can coexist with negative mean alpha: fitted factor contributions can exceed mean dependent excess returns. Fit describes sample variation; alpha describes a conditional mean residual. Neither independently establishes investable alpha.

All nominally significant primary intercepts are negative. No intercept survives supplementary Holm adjustment within models or across 615 primary tests. Non-rejection does not prove exactly zero alpha, efficiency, or equivalent models. HAC lag sensitivities from the accepted audit retain their conditional interpretation and do not resolve H1.

Test portfolios share constituents and contain six duplicated return-series pairs; errors and exposures are dependent. A self-inclusion sensitivity retains 109 assets after excluding 14 directly related assets. It is a qualified diagnostic, not an independent market sample. GRS and spanning results remain unavailable.

VIII Portfolio construction costs and risk

The eight existing methods are equal weight, minimum variance, mean variance, maximum Sharpe, maximum diversification, risk parity, hierarchical risk parity, and CVaR. They allocate to 123 long-only quantile sleeves, not directly to 487 equities or long-short spreads. Sleeve returns omit a complete constituent-level execution model; outer allocations are not independently executable institutional strategies.

The original 36-cycle rolling history, long-only fully invested unlevered weights, covariance conventions, objectives and bounds are unchanged. No position or turnover cap was added. Phase 3 checked feasibility and objective identities, not global optimality of every nonconvex objective. Optimizer cash remains configured at zero, distinct from interval-matched cash in realized performance.

Optimizer arrays exclude evaluated future rows, but some last historical marks coincide with modeled holding start. Exit-labeled records and UTC-day availability do not resolve every intraday close-order boundary. Portfolio timing remains source-blocked; allocations use modeled closes, not verified orders.

Table VIII Modeled transaction cost components in basis points

Component	LOW	BASE	HIGH
Full quoted spread	4	10	20
Charged half-spread	2	5	10
Slippage component	1	2	4
Fixed allowance in slippage	2	3	6
Commission	0	0	0
Total per absolute traded weight	5	10	20

Note. Quoted spread is not itself charged: half-spread is charged. Fixed allowance is included in total modeled slippage, not added twice. Nonlinear impact is zero; assumptions are not order-book calibration.

L1 is summed absolute target-minus-pretrade weights; one-way turnover is half L1. Fee equals L1 times the per-side rate. Initial deployment has L1 = 1 and costs 10 bps in BASE. A full invested switch has L1 = 2 and costs 20 bps at the same rate; this is not double counting. Net cycle return is gross less cost. Pretrade weights drift with gross returns under the existing convention. Equal weight and optimized allocations share cash starts, schedules and accounting.

Table IX Corrected BASE portfolio performance

Method	Ann net	Volatility	Sharpe	MDD	Cum net
Equal weight	15.87%	10.74%	1.023	-8.29%	46.31%
Minimum variance	15.50%	9.95%	1.066	-6.50%	45.09%
Mean variance	10.45%	14.65%	0.456	-16.11%	29.28%
Maximum Sharpe	21.83%	12.05%	1.346	-6.16%	66.56%
Maximum diversification	19.64%	10.64%	1.334	-4.83%	58.91%
Risk parity	15.60%	10.43%	1.028	-7.99%	45.42%
Hierarchical risk parity	15.11%	10.19%	1.008	-7.72%	43.84%
CVaR	23.83%	14.49%	1.252	-7.08%	73.68%

Note. 31 complete cycles, exit labels 1 February 2024-3 August 2026, from January 2024-July 2026 signals. Annualization uses 12 periods/year. Net returns include outer fees, not complete constituent execution costs.

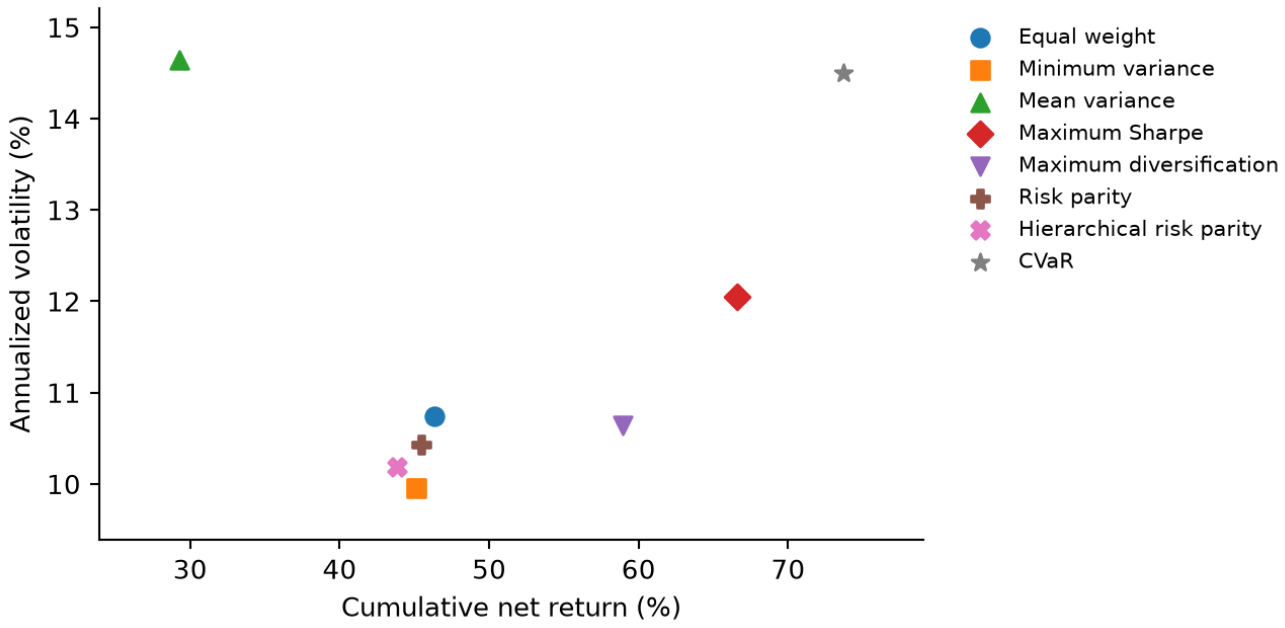


Figure 4. Corrected BASE cumulative net return versus annualized volatility. Descriptive positions do not establish persistent superiority.

Table X Corrected cumulative return sensitivity to modeled costs

Method	Gross	LOW net	BASE net	HIGH net
Equal weight	46.53%	46.42%	46.31%	46.10%
Minimum variance	46.34%	45.72%	45.09%	43.86%
Mean variance	32.81%	31.03%	29.28%	25.83%
Maximum Sharpe	68.70%	67.63%	66.56%	64.44%
Maximum diversification	60.16%	59.53%	58.91%	57.67%
Risk parity	45.67%	45.55%	45.42%	45.18%
Hierarchical risk parity	44.43%	44.14%	43.84%	43.25%
CVaR	74.83%	74.26%	73.68%	72.55%

Note. Within a method, scenarios share the same weight path. Compounded fee effects are not simple sums of rates.

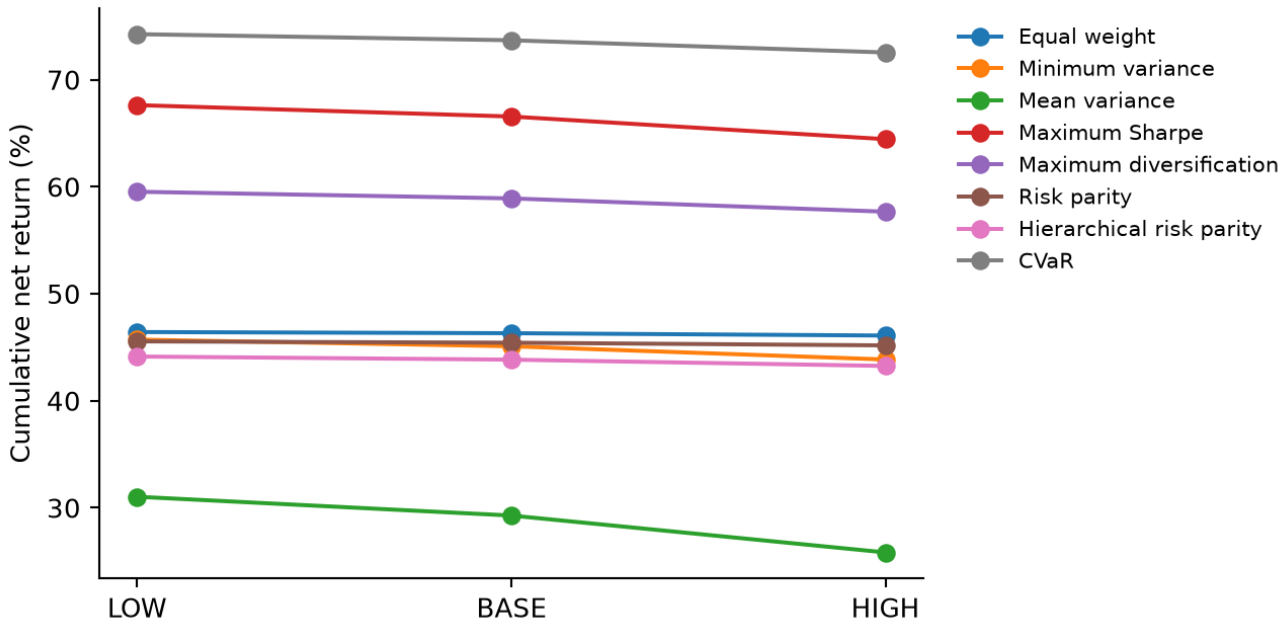


Figure 5. Corrected LOW, BASE and HIGH cumulative net returns. Weight paths are invariant and larger fees reduce all eight cumulative outcomes.

Corrected CVaR gives 73.68% cumulative net return and Sharpe 1.252 over 31 cycles. Historical 32-cycle values, 83.17% and 1.667, reconcile on old inputs. Cash alignment alone on the unchanged old path gives Sharpe 1.379. Remaining changes involve holding inputs and endpoints, not fee changes or tuning. The historical maximum-diversification drawdown -5.43% is valid on its old path; the corrected-path value is -4.83%. Including initial NAV changes none of the eight old drawdown minima.

Table XI Corrected monthly cycle loss risk

Method	VaR95	Tail mean loss95	Tail count
Equal weight	4.66%	5.30%	2
Minimum variance	4.34%	5.90%	2
Mean variance	5.09%	8.66%	2
Maximum Sharpe	4.82%	5.08%	2
Maximum diversification	3.91%	4.79%	2
Risk parity	4.61%	5.32%	2
Hierarchical risk parity	4.57%	5.37%	2
CVaR	4.95%	6.63%	2

Note. Positive loss magnitudes, 95% confidence, unannualized cycle returns. VaR uses the 5th return percentile; tail mean includes returns at or below it. This two-observation tail differs from fractional-tail CVaR optimization on 36 scenarios.

One paired diagnostic summarizes CVaR minus equal-weight net cycle returns: mean 0.005984 decimal, with leave-one-cycle-out means 0.004087-0.008760. This range is descriptive, not a confidence interval or superiority test. Thirty-one cycles and sleeve dependence restrict generalization.

Table XII Historical scenario coverage

Family	Revision treatment
Market	Exposure mapping only; no corrected forecast or execution claim
Rates	Exposure mapping only; no corrected forecast or execution claim
Volatility	Exposure mapping only; no corrected forecast or execution claim
Inflation	Exposure mapping only; no corrected forecast or execution claim
Liquidity	Exposure mapping only; no corrected forecast or execution claim

Note. Original instantaneous impacts are not propagated to the corrected experiment. The custom sixth mapping lacks a stated interpretation in the supplied paper; no guessed replacement is added.

Scenario mappings are not return forecasts and were outside the corrected performance reconciliation. Their superseded impacts are omitted. Old-path tracking error, information ratio, Sortino and beta are not silently copied into corrected tables; the claim ledger records their omission.

IX Walk forward machine learning

Book-to-market and 12-minus-1 momentum remain the two features. ML consumes frozen raw scores, not corrected quantile returns or pricing coefficients. Its historical portfolio parent supplies only fee context; keeping that parent correctly reflects dependence. The target is security minus SPY compounded return over 21 common observed sessions after decision. Complete endpoints are required; incomplete histories are not extended to a security's next 21 quotes. The return basis is frozen adjusted quotes, not certified dividend-inclusive returns.

Historical 27,640 rows and 91 distinct dates represent only 89 months: two earlier last-security observations created false date slots. Correction leaves 27,632 rows, 89 common month ends, 15 complete folds and 7,286 held-out observations per model. Two off-month-end rows and six incomplete labels are excluded, with no invented, forward-filled or cash-replaced observations.

Training expands from 60 to 74 calendar months across folds, rather than rolling at a fixed 60. Validation uses 12 months and the existing one-month embargo is retained at both boundaries. Actual label intervals do not cross subsequent formation boundaries. First training is February 2019-January 2024, February is omitted, validation is March 2024-February 2025, March 2025 is omitted, and first test is 30 April 2025. Last test is 30 June 2026. Labels overlap in five of 14 adjacent test pairs: folds are dependent.

Preprocessors use permitted training data. Recorded-day gates and final fitted scalers pass; baseline, linear and Ridge predictions were independently reproduced. Retrospective predictions do not authenticate historical live forecasts. Exhaustive filing capture and same-close availability remain source-blocked; fully leakage-free live execution is not claimed.

The eight named families are zero, historical mean, factor composite, linear, Ridge, Elastic Net, random forest and XGBoost. Factor composite and linear instantiate the same LinearRegression estimator and yield identical vectors and ranks: aliases, not independent confirmations. Ridge final alpha is 0.1; its weak penalty relative to training information explains close forecasts. Search grids, seed 17, one thread and validation-IC selection remain unchanged (Appendix G).

A controlled refit was necessary because calendar/label defects changed the fitting population. The later constant-correlation repair republished metrics with identical fitted-model/prediction bytes. No family or parameter was added to improve outcomes. Mean IC is the unweighted mean of valid date-level cross-sectional Spearman correlations; constant forecasts remain undefined.

The corrected bootstrap uses moving blocks of three date ICs, 100 replicates and seed 17; its point estimate is the observed date mean. The original 1,000-independent-resample description did not match implementation. Few dependent folds, overlap, model multiplicity and the small resampling budget make intervals exploratory rather than evidence of stable predictability.

Table XIII Corrected held out forecast results

Model	Mean IC	Median IC	Exploratory 95% interval
Zero	Undefined	Undefined	Undefined
Historical mean	Undefined	Undefined	Undefined
Factor composite	0.034430	0.016341	[0.010731, 0.084270]
Linear	0.034430	0.016341	[0.010731, 0.084270]
Ridge	0.034434	0.016336	[0.010735, 0.084272]
Elastic Net	0.044609	0.059116	[0.016324, 0.096766]
Random forest	-0.002641	-0.011912	[-0.017721, 0.015985]
XGBoost	-0.023285	-0.041750	[-0.038821, -0.005015]

Note. 15 folds and 7,286 held-out observations per family. Constant baselines have no valid IC. Intervals are not adjusted for eight named comparisons.

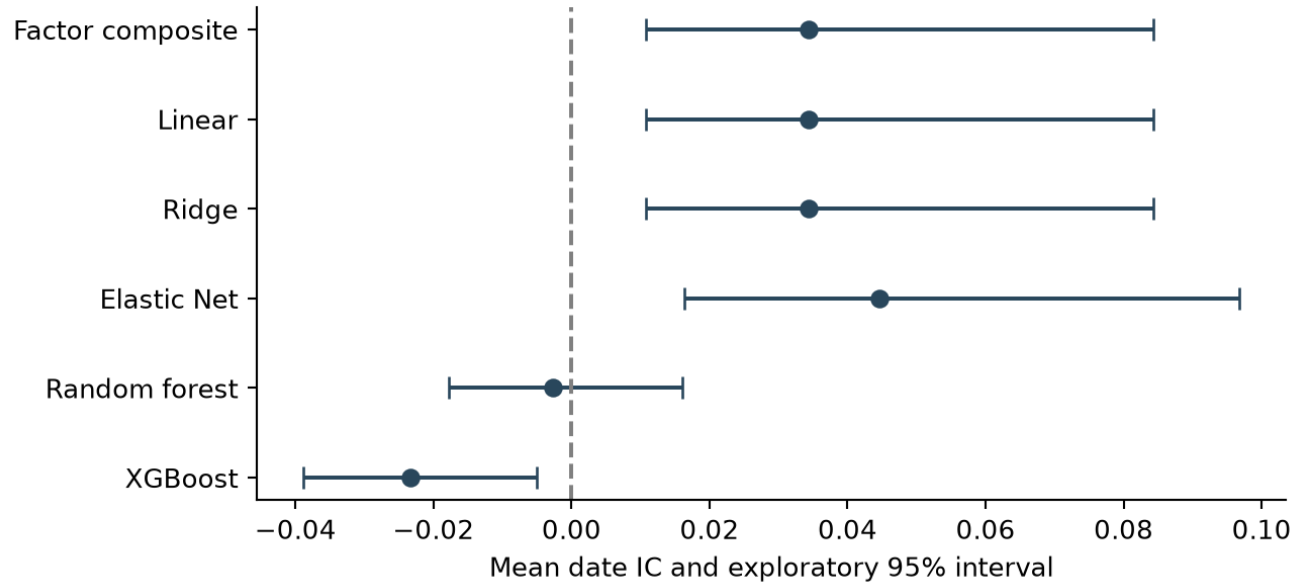


Figure 6. Corrected mean date IC and exploratory 95% block intervals. Constant baselines are undefined and omitted; factor composite and linear coincide.

The claim that all intervals include zero is superseded. Linear-family intervals are nominally positive; random forest crosses zero; XGBoost is nominally negative. These are forecast statements, not stable alpha. Historical linear predictions restricted to corrected test keys give IC about 0.037238, above corrected linear IC 0.034430. Whole-sample old/new comparisons do not isolate improvement.

Table XIV Corrected one session economic checks

Model	Mean net return	Positive folds	Initial fee
Zero	-0.0604%	7/15	0.10%
Historical mean	-0.0604%	7/15	0.10%
Factor composite	0.0559%	7/15	0.10%
Linear	0.0559%	7/15	0.10%
Ridge	0.0559%	7/15	0.10%
Elastic Net	0.0551%	7/15	0.10%
Random forest	0.1410%	8/15	0.10%
XGBoost	0.1554%	10/15	0.10%

Note. Each fold resets to cash and unit NAV, selects the highest-ranked 20% with equal weights and holds for one subsequent common session. These are not 21-session returns or compounded wealth.

The economic evaluator charges initial deployment, not full entry-to-liquidation costs or a continuous capital chain. Constant ties use deterministic security ordering and provide no forecasting signal. Held-out rankings determine membership without using future outcomes, but historical-clock limits remain. Stable incremental after-cost economic value has not been established despite some positive one-session means.

Permutation importance measures validation-loss changes; SHAP explains test predictions with a historical background. Their targets and samples differ. Top-feature disagreement occurs in seven of 15 random-forest folds and six of 15 XGBoost folds; training feature correlation is about -0.1875. Correlation is not shown to cause the disagreement. Attributions are associations, not causal investment effects.

X Robustness and research integrity checks

Accepted evidence includes holding calendars, cost sensitivities, multiple allocation methods, purging, embargo, bootstrap summaries, feature ablations and publication authentication. Each addresses a bounded question. Old scenario and restart checks do not automatically validate new empirical claims.

The final Phase 3 suite passed 372 tests with 90.62% coverage, including a focused 91-test portfolio/ML run. Tests verify behavior over tested cases and complement frozen-output reconciliations. They do not prove causality, representativeness, dividends or execution. This revision does not rerun regressions, portfolio publications, model training, SEC ingestion or factor construction for typesetting.

XI Hypothesis assessment

Table XV Revised hypothesis interpretation

ID	Assessment	Evidence boundary
H1	Inconclusive	Original cost/multiplicity decision missing
H2	Partially supported descriptively	Higher in-sample fit; stable economic extension unestablished
H3	Inconclusive	Original direction/approval and stability decision missing
H4	Supported in fixed simulation	Monotonic fixed-weight fee sensitivity is a simulation identity
H5	Partially supported descriptively	Different short-sample profiles; durable superiority unestablished
H6	Stable joint proposition unestablished	Some nominal IC; stable joint after-cost value unestablished
H7	Not supported descriptively	Attributions differ across methods, samples and targets

Note. Historical propositions are preserved. These labels summarize bounded evidence, not newly authenticated confirmatory rules. Appendix I gives complete wording and evidence mapping.

XII Discussion

Corrections materially change historical-table interpretation. Mean fit is higher and mean cycle alpha is negative; aligned intervals and samples do not constitute newly discovered pricing power. Shared constituents, duplicate series and correlated residuals constrain what fit establishes.

Allocation objectives alter observed return and risk. CVaR has highest cumulative return, while maximum Sharpe has highest corrected BASE Sharpe. Neither is generally superior. Short horizons, sleeve costs and timing limits prevent an institutional-investability conclusion. Cost monotonicity supports H4 inside the fixed model, not empirical implementation shortfall or capacity calibration.

ML forecast evidence differs from economic evidence. Some nominal intervals exclude zero and XGBoost is negative, so the uniformly-zero-crossing narrative is inaccurate. Fifteen dependent dates, model multiplicity, a small bootstrap and one-session economics prevent robust-alpha claims. The contribution is integration, traceable accounting and conditional inference with limitations attached to affected claims.

XIII Limitations

Data and identity. The panel is not CRSP-, Compustat-, S&P 500- or Russell-replication evidence. Sparse early cross-sections and survivorship selection constrain generalization. Share-class mapping, amendments and exhaustive SEC capture chronology remain incomplete. Current documentation does not certify frozen source vintage.

Returns and execution. Adjusted quotes do not establish complete dividend-inclusive total returns. Delayed closes are modeled, not verified fills; intraday decision and receipt boundaries remain unresolved. Missing endpoints are excluded rather than repaired.

Inference. Names are not independent factors. Test portfolios share securities and duplicate series; q proxies are study specific. High conditional fit establishes neither causal pricing nor exactly zero alpha. H1/H3 approval/decision evidence remains absent.

Investment. Thirty-one cycles cannot establish durable rankings. Costs are assumptions; constituent turnover, capacity and complete execution remain unverified. Tail losses use two observations. The paired leave-one-out range is descriptive.

Prediction. Forecasts are retrospective. Fifteen dependent folds, overlapping labels, multiple models and 100 bootstrap replicates constrain interval interpretation. Linear and factor composite coincide. One-session cash-reset economics does not measure a continuous 21-session strategy.

XIV Future research

Future extensions could use a point-in-time delisting-inclusive master, independent corporate-action evidence, consolidated volume comparisons, longer evaluation and observed execution where lawfully available. Stronger ML inference needs more independent evaluation information and a specified investment horizon. These are proposed extensions, not completed robustness tests.

XV Conclusion

An integrated public-data-oriented workflow connects security evidence, characteristics, matched pricing comparisons, allocations and temporally ordered forecasts. Corrected fit is high, mean intercepts are negative and no primary intercept survives supplementary Holm adjustment. Portfolio rules differ descriptively; CVaR returns 73.68% over 31 complete cycles. Some models have nominal positive forecast intervals; XGBoost is nominally negative. Neither persistent portfolio superiority nor stable incremental after-cost ML value is established. Missing source-vintage, timing and original hypothesis-decision evidence remains part of the result.

Data and code availability

Repository: https://github.com/amitdudi04/institutional-multi-factor-asset-pricing-platform. Original tag project-complete-public-data-v1 and commit 5a3e930665756fa7aaedeceb5f9ab90792bcf849 identify the historical release, not corrected findings. This revision uses accepted Phase 3 baseline f22f5cbe365e98c8d5fef6b5308ff765703df173 and revision IMFP-P4-2026-11-01. The containing Phase 4 evidence commit and checksum manifest identify delivery; no self-referential hash is embedded.

Code, configuration and audit reconciliations are public. Frozen Parquet and model bundles are not asserted public; exact reproduction needs lawful original bytes. Current downloads are not equivalent. Software licensing does not grant third-party-data rights. Hashes authenticate identity but do not supply absent inputs or metadata. Appendix H identifies accepted final parents.

FUNDING

This research received no external funding.

COMPETING INTERESTS

The author declares no competing interests.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES

AI-assisted tools were used for limited technical and editorial support, including coding assistance, debugging, document formatting, and language refinement. The research questions, data-source selection, empirical methodology, model specifications, validation procedures, analysis, interpretation of results, and conclusions were determined and verified by the author. The author reviewed the generated outputs and takes full responsibility for the final manuscript.

ACKNOWLEDGMENT

This research was conducted under the academic supervision of Mitul Sudhirkumar Nagar, Assistant Professor, Department of Computational Intelligence, SRM Institute of Science and Technology.

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Appendix A Complete factor definition catalog

The catalog retains the 48 configured IDs and their conceptual formulas. Formula labels containing total return are historical registry terminology; the empirical basis remains frozen provider-vintage adjusted quotes with incomplete dividend certification. Definitions overlap economically. Source: versioned factors/definitions.py and factors.yaml, bound in the publication source manifest.

Table A1 Configured definitions

ID	Inputs	Conceptual formula
market_return	market_return	broad-market adjusted-quote return proxy
excess_return	market_return, risk_free	market return proxy - risk-free return
risk_free_rate	risk_free	frequency-matched approved risk-free simple return
rolling_market_beta	return, market_return	rolling cov(return, market)/var(market)
market_cap	price, shares_outstanding	price * shares outstanding
log_market_cap	price, shares_outstanding	ln(market capitalization)
size_small	market_cap, exchange	market cap <= NYSE 30th percentile
size_mid	market_cap, exchange	NYSE 30th < market cap < NYSE 70th
size_large	market_cap, exchange	market cap >= NYSE 70th percentile
book_to_market	book_equity, market_cap	book equity / market capitalization
price_to_book	book_equity, market_cap	market capitalization / book equity
earnings_yield	net_income, market_cap	net income / market capitalization
cash_flow_yield	operating_cash_flow, market_cap	operating cash flow / market capitalization
dividend_yield	dividends, market_cap	dividends / market capitalization
momentum_1m	return	compound return over 21 sessions
momentum_3m	return	compound return over 63 sessions
momentum_6m	return	compound return over 126 sessions
momentum_12_1m	return	compound return months 12 through 2
momentum_12m	return	compound return over 252 sessions
momentum_24m	return	compound return over 504 sessions
residual_momentum	return, market_return	compound residual return after rolling market beta
roe	net_income, shareholder_equity	net income / shareholder equity
roa	net_income, total_assets	net income / total assets
gross_profitability	gross_profit, total_assets	gross profit / total assets
operating_margin	operating_income, revenue	operating income / revenue
net_margin	net_income, revenue	net income / revenue
asset_turnover	revenue, average_assets	revenue / average assets
accruals	total_accruals, total_assets	total accruals / total assets

ID	Inputs	Conceptual formula
leverage	total_debt, total_assets	total debt / total assets
debt_to_equity	total_debt, shareholder_equity	total debt / shareholder equity
interest_coverage	operating_income, interest_expense	operating income / interest expense
asset_growth	total_assets, prior_total_assets	total assets / prior total assets - 1
capex_growth	capex, prior_capex	capital expenditure / prior capital expenditure - 1
equity_issuance	net_equity_issuance, shareholder_equity	net equity issuance / shareholder equity
working_capital_growth	working_capital, prior_working_capital	working capital / prior working capital - 1
rolling_volatility	return	annualized rolling standard deviation
downside_volatility	return	annualized rolling standard deviation of negative returns
semi_variance	return	annualized mean squared negative return
beta	return, market_return	rolling market beta
idiosyncratic_volatility	return, market_return	annualized volatility of market residual
average_dollar_volume	price, volume	rolling mean price * volume
turnover	volume, shares_outstanding	volume / shares outstanding
amihud_illiquidity	return, price, volume	abs(return) / dollar volume
bid_ask_proxy	high, low	(high-low) / midpoint
rolling_beta	return, market_return	rolling market beta
rolling_correlation	return, market_return	rolling correlation with market
tracking_error	return, benchmark_return	annualized volatility of return-benchmark
downside_beta	return, market_return	cov(return, market)/var(market) on negative-market dates

Note. equity_issuance has no estimable raw values. The table states definitions, not independent factor discoveries; names with beta or size mappings can duplicate exposures.

Appendix B Screening and sample windows

Table B1 Historical screening reasons

Reason	Rejected	Share of 822
Attribution	114	13.87%
Source splice	56	6.81%
Identity	68	8.27%
Shares	70	8.52%
Terminal event	16	1.95%
Split basis	7	0.85%
Asset type	2	0.24%
Non-US or venue	1	0.12%
Fundamentals	1	0.12%
Total	335	40.75%

Note. Historical screening tallies retained from the archived study document and evidence ledger; classifications describe exclusions and do not certify absent provider securities.

Table B2 Corrected analytical windows

Stage	Window or count	Meaning
Market	2018-02-14 to 2026-08-04	Frozen daily observations
Fundamentals	2019-02-23 to 2026-08-15	Recorded availability, not decision sample
Factor portfolios	11,352 rows; 123 series	Complete delayed-close cycles; starts differ by characteristic
Primary pricing	2020-04-01 to 2026-08-03 exits	67-77 matched cycles per asset
Portfolio evaluation	2024-02-01 to 2026-08-03 exits	31 complete cycles
ML tests	2025-04-30 to 2026-06-30	15 common monthly test dates

Appendix C Additional descriptive statistics

Table C1 Frozen input quantiles

Variable	P1	P25	Median	P75	P99
security return	-0.070171	-0.010529	0.000575	0.011570	0.073402
market capitalization	4.9231e+08	1.0146e+10	2.1601e+10	5.0167e+10	3.6731e+11
daily dollar volume	153266.120685	4.8970e+06	1.2678e+07	4.2316e+07	6.6719e+08
book to market	-0.201907	0.118791	0.292552	0.557899	1.982315
momentum 12 1m	-0.601507	-0.118453	0.069121	0.283342	1.645525
rolling volatility	0.157417	0.252073	0.329544	0.462254	1.167571

Note. Units and N are in Table IV. Sample SD uses n-1. Quantiles are direct tabulations of observed nonmissing values, not additional significance tests.

Appendix D Asset pricing inference notes

The primary pricing publication is asset-pricing-d60bee7310b34976c80372a0f0f29e21. The separate model-specific supplementary publication is asset-pricing-8c2396179c4b509796b2cab043ec31d2. Both descend from corrected factor publication factor-4ddee9ac6fb45cedfd62a2cc68676b58. Model-specific and historical statistics must not be substituted into the matched-sample Table VII.

The accepted audit independently reproduced 615 primary and 615 supplementary fits, including adjusted R-squared and HAC arithmetic. Three lags define primary inference; supplementary lag sensitivities and Holm adjustments qualify interpretation. Intercept estimates and standard errors are decimal cycle returns. The high fit, shared construction, and negative mean intercepts are consistent conditional findings, not independent tests of a market-wide theory.

Table D1 Additional primary summary statistics

Model	Median adj R^2	Median alpha	Mean alpha SE
CAPM	0.787564	-0.001714	0.003147
FF3	0.822721	-0.002071	0.002779
Carhart 4	0.830492	-0.001978	0.002720
FF5	0.835738	-0.002123	0.002809
q-style	0.828120	-0.002437	0.002996

Note. SE and alpha are decimal per monthly trading cycle. This descriptive summary is not a joint pricing-error test.

Appendix E Portfolio and risk evidence coverage

The final correction preserves eight methods and three modeled fee scenarios, producing 24 immutable portfolio publications. Independent reconciliation covers genuine net-return paths, actual traded weights, gross drift, fee identities, objective values, feasibility, and risk calculations. Historical paths remain archived. No raw historical CSV is overwritten.

Table E1 Historical and corrected BASE comparison

Method	Old cum net	New cum net	Old Sharpe	New Sharpe
Equal weight	48.89%	46.31%	1.383	1.023
Minimum variance	41.64%	45.09%	1.341	1.066
Mean variance	37.60%	29.28%	0.818	0.456
Maximum Sharpe	64.61%	66.56%	1.575	1.346
Maximum diversification	53.08%	58.91%	1.499	1.334
Risk parity	47.72%	45.42%	1.397	1.028
Hierarchical risk parity	46.79%	43.84%	1.403	1.008
CVaR	83.17%	73.68%	1.667	1.252

Note. Old: 32 cycles with historical endpoints and daily-cash mismatch. New: 31 full cycles and interval cash. This is not a matched-horizon causal treatment effect.

The 95% historical loss tail in Table XI averages two cycle observations. It is not annualized and differs from the CVaR optimizer's fractional-tail scenario objective. The omitted legacy scenario, tracking, information-ratio,

beta and Sortino cells are explicitly recorded in the revision ledger; their absence does not mean that their historical values were zero.

Appendix F Transaction costs and worked calculations

Table F1 Transparent arithmetic

Calculation	Result
Acceptance	487 / 822 = 59.25%
Rejection	335 / 822 = 40.75%
Corrected FF5 minus CAPM fit	0.056840
Equal-weight BASE wealth drag	0.2178 percentage points
Mean-variance HIGH wealth drag	6.9845 percentage points
Initial BASE deployment	1 × 0.001 = 0.001 NAV
Full invested BASE switch	2 × 0.001 = 0.002 NAV

Note. Fee examples are accounting identities; only publication-derived wealth drags are empirical summaries.

Appendix G ML configuration and evaluation

Table G1 Original candidate grids retained

Family	Grid
Zero and historical mean	Constant 0 or permitted training-target mean
Factor composite and linear	LinearRegression; same feature ordering
Ridge	alpha: 0.1, 1.0, 10.0
Elastic Net	alpha: 0.0001, 0.001; l1_ratio: 0.25, 0.5, 0.75
Random forest	n_estimators: 50, 100; max_depth: 3, 6
XGBoost	n_estimators: 50, 100; max_depth: 2, 3; learning_rate: 0.05, 0.1

Note. Seed 17, one thread, training-only fitted transforms, validation Spearman selection. Test returns are not tuning inputs. No family or grid was expanded.

The full forecast reconciliation includes date IC, median IC, RMSE, constant-baseline undefined cases, block intervals and one-session net economic means. Final serialized corrected models reproduce final-fold forecasts. Earlier tree fits were not all independently retained; archived predictions and provenance do not imply that every historical tree was independently refitted.

Appendix H Reproducibility identifiers and lineage

Table H1 Upstream authority

Evidence	Identifier
Market input	factor_market_input-89848f8e3c34d53d0432f6e0
Fundamental input	factor_fundamental_input-962c8ada3e74e32b00568cd7
Historical factor	factor-5a73616f131fca2102fa2ca4141ad64f
Historical pricing	asset-pricing-c0a72fae5349f1048e18557250da1927
Corrected factor	factor-4ddee9ac6fb45cedfd62a2cc68676b58
Primary pricing	asset-pricing-d60bee7310b34976c80372a0f0f29e21
Supplementary pricing	asset-pricing-8c2396179c4b509796b2cab043ec31d2

Table H2 Corrected BASE portfolio identities

Method	Immutable publication ID
Equal weight	portfolio-56ff12e1590325ab00bdd65e67bdc83e
Minimum variance	portfolio-da8cd6b77d46e57138a924a8c9d106eb
Mean variance	portfolio-99d63f9cb5aecc834329df585ebc2577
Maximum Sharpe	portfolio-bb374ca488687ec7e964587f30e3bd15
Maximum diversification	portfolio-af2465790ec1d893a2f71af83b3cfc4d
Risk parity	portfolio-102f3597c9fdfe7e17e7e910ec1173cd
Hierarchical risk parity	portfolio-66f902bdd7cf1bfb2b8c250de9717e8f
CVaR	portfolio-87fa6e8666cac548ba3c936ef55cb733

Note. All LOW/HIGH IDs, configuration hashes, parent hashes and source commits are enumerated in PHASE4_PUBLICATION_SOURCE_MANIFEST.json; none is regenerated for this revision.

Table H3 Corrected final ML identities

Family	Immutable publication ID
Zero	ml-6731aae58077b43bafcb27e2ede9b879
Historical mean	ml-8465c40774367a85a0d73163324ba5c6
Factor composite	ml-7d6d0a56ed4aef365ee6510c4bd28b81
Linear	ml-b7855cf2b3ddbe94b38d20a544743469
Ridge	ml-0dce7f82c4a92e971bb4740a7d0d3632
Elastic Net	ml-d35e503f6a33010aa0346e2eda1b8678
Random forest	ml-b2185fbf1e37089aeb1c0b122e757890
XGBoost	ml-74e283ab3be2bba30aa7d6d21281951f

Note. These are the final metric-corrected publications, not superseded intermediate IDs. Fitting source is 0fe382f969197928af35d15cc619d9c2dc6105f8; metric-only source is 9df5455 as fully expanded in the bound manifest.

Portfolio publications depend on corrected quantile returns; pricing is contextual. ML keeps the actual historical raw-score and market parents and historical fee context because it does not consume corrected quantile returns or pricing coefficients. Hashes and source commits distinguish fitted-model production from metric-only

republication. The external Phase 4 source manifest binds all 32 final authorities and manuscript hashes; the 143-publication runtime count from the historical paper is not presented as the corrected research inventory.

Appendix I Hypothesis evidence and original wording

H1 Historical proposition and evidence

Original supplied wording: At least one factor spread survives the complete pre-specified cost and multiplicity rule. Intended evidence: Complete original spread-level net-cost and multiplicity decision. Historical conclusion: Inconclusive. Revised interpretation: Inconclusive. Original cost/multiplicity decision missing.

Evidence: Phase 2 AP10 provenance record and unchanged Phase 3 limitations. Remaining limitation: Original approval/decision record is missing; not reconstructed after observing results.

Citation: outputs/audit/phase2/PHASE2_ASSET_PRICING_VALIDATION.md#ap10.

H2 Historical proposition and evidence

Original supplied wording: Multifactor models improve explanatory fit relative to CAPM and that improvement extends to stable out-of-sample economic evidence. Intended evidence: Matched model fit plus stable OOS economic extension. Historical conclusion: Partially supported. Revised interpretation: Partially supported descriptively. Higher in-sample fit; stable economic extension unestablished.

Evidence: Phase 2C primary model summary and Phase 3 portfolio/ML reconciliations. Remaining limitation: Descriptive evidence and dependent selected samples do not establish the stronger general proposition.

Citation: outputs/audit/phase3/PHASE3_COMPLETE_VALIDATION.md.

H3 Historical proposition and evidence

Original supplied wording: Estimated relations are stable through time under the original rule. Intended evidence: Original rolling-relation stability rule and approved direction. Historical conclusion: Inconclusive. Revised interpretation: Inconclusive. Original direction/approval and stability decision missing.

Evidence: Phase 2 AP10 provenance record and unchanged Phase 3 limitations. Remaining limitation: Original approval/decision record is missing; not reconstructed after observing results.

Citation: outputs/audit/phase2/PHASE2_ASSET_PRICING_VALIDATION.md#ap10.

H4 Historical proposition and evidence

Original supplied wording: Higher transaction-cost assumptions reduce cumulative outcomes. Intended evidence: Same-weight LOW/BASE/HIGH cumulative comparisons. Historical conclusion: Supported. Revised interpretation: Supported in fixed simulation. Monotonic fixed-weight fee sensitivity is a simulation identity.

Evidence: Phase 2C primary model summary and Phase 3 portfolio/ML reconciliations. Remaining limitation: Descriptive evidence and dependent selected samples do not establish the stronger general proposition.

Citation: outputs/audit/phase3/PHASE3_COMPLETE_VALIDATION.md.

H5 Historical proposition and evidence

Original supplied wording: Constrained methods alter risk-return trade-offs and any apparent superiority is stable out of sample. Intended evidence: Risk/return comparison plus stable OOS superiority. Historical conclusion:

Partially supported. Revised interpretation: Partially supported descriptively. Different short-sample profiles; durable superiority unestablished.

Evidence: Phase 2C primary model summary and Phase 3 portfolio/ML reconciliations. Remaining limitation: Descriptive evidence and dependent selected samples do not establish the stronger general proposition.

Citation: outputs/audit/phase3/PHASE3_COMPLETE_VALIDATION.md.

H6 Historical proposition and evidence

Original supplied wording: ML adds stable OOS predictive and after-cost value. Intended evidence: Temporal held-out IC and after-cost economic evidence. Historical conclusion: Not supported; all IC intervals claimed to cross zero. Revised interpretation: Stable joint proposition unestablished. Some nominal IC; stable joint after-cost value unestablished.

Evidence: Phase 2C primary model summary and Phase 3 portfolio/ML reconciliations. Remaining limitation: Descriptive evidence and dependent selected samples do not establish the stronger general proposition.

Citation: outputs/audit/phase3/PHASE3_COMPLETE_VALIDATION.md.

H7 Historical proposition and evidence

Original supplied wording: Explanation rankings are stable across methods. Intended evidence: Comparable explanation ranking stability. Historical conclusion: Not supported. Revised interpretation: Not supported descriptively. Attributions differ across methods, samples and targets.

Evidence: Phase 2C primary model summary and Phase 3 portfolio/ML reconciliations. Remaining limitation: Descriptive evidence and dependent selected samples do not establish the stronger general proposition.

Citation: outputs/audit/phase3/PHASE3_COMPLETE_VALIDATION.md.

Version comparison: the older 30-page PDF states H2 as "Multifactor models improve explanatory fit relative to CAPM" and H5 as "Constrained methods alter risk-return trade-offs"; the supplied 14-page SSRN text adds stable out-of-sample extensions. These are documented historical differences, not retroactive changes made here. The accepted audit also identifies an earlier H3 direction mismatch. The author must identify the approved version before claiming original confirmatory hypothesis decisions.

Appendix J Software assurance and empirical validation

Accepted Phase 3 records 372 passing full-suite tests, 105 warnings and 90.62% coverage, plus 91 targeted tests and six new regressions. One preliminary broad run was interrupted when a further degenerate-correlation defect was found; one final completed run follows the correction. Type, lint, format and lock checks passed at that checkpoint. This revision changes documents and evidence only and does not claim a new full-suite execution.

Software assurance, artifact authentication, independent numerical reconciliation and source-data validity are different layers. Tests and hashes cannot establish historical receipt times, investment capacity, dividend treatment or original hypothesis approval. Public audit evidence supports bounded reproduction; restricted inputs and unresolved upstream provenance remain explicit.

Appendix K Formula reference

The following editable equations retain the historical methods and clarify the implemented conventions. R is a simple decimal return, L is loss, w is target weight, w− is pretrade weight, Σ is covariance, μ is mean scenario return, n is sample size and k is the number of regressors excluding the intercept. K is 12 for reported cycle annualization. Algebraic full-rank OLS identities do not imply explicit matrix inversion in software.

$$Rank(i,t) = rank(X(i,t)) / N(t) \quad (1)$$

A descriptive rank identity; actual score normalization follows the released robust preprocessing, with tied ranks governed by implementation.

$$F(t) = R(high,t) - R(low,t) \quad (2)$$

Factor spreads differ from the long-only quantile sleeves used by allocation.

$$R(p,t) - R(f,t) = \alpha(p) + \beta(M,p)[R(M,t) - R(f,t)] + \varepsilon(p,t) \quad (3)$$

CAPM, aligned cycle excess returns.

$$R(p,t) - R(f,t) = \alpha + \beta(M)MKT + \beta(S)SMB + \beta(H)HML + \varepsilon \quad (4)$$

FF3 uses internal factor proxies.

$$Carhart4: FF3\ right\ side + \beta(MOM)MOM \quad (5)$$

Momentum extends the FF3 regressors.

$$FF5: \alpha + \beta(M)MKT + \beta(S)SMB + \beta(H)HML + \beta(R)RMW + \beta(C)CMA + \varepsilon \quad (6)$$

The dependent variable remains portfolio excess return.

$$q-style: R(p,t) - R(f,t) = \alpha + \beta(M)MKT + \beta(ME)ME + \beta(IA)IA + \beta(ROE)ROEproxy + \varepsilon \quad (7)$$

ME derives from log cap; IA from asset growth; ROEproxy from earnings yield.

$$\beta = (X^{T}X)^{-1}X^{T}y \quad (8)$$

Full-rank OLS algebra; HAC changes covariance, not point estimates.

$$R^2adj = 1 - (1 - R^2)(n - 1)/(n - k - 1) \quad (9)$$

Use model-specific k and matched n.

$$min(w)\ w^{T}\Sigma w \quad (10)$$

Minimum variance, subject to unchanged constraints.

$$max(w)\ w^{T}\mu - (\lambda/2)w^{T}\Sigma w \quad (11)$$

Mean-variance objective; original λ retained.

$$max(w)\ [w^{T}\mu - Rf]/\sqrt{(w^{T}\Sigma w)} \quad (12)$$

Maximum Sharpe optimizer uses its original configured cash assumption.

$$DR(w) = w^{T}\sigma / \sqrt{(w^{T}\Sigma w)} \quad (13)$$

Maximum diversification uses standalone volatilities σ.

$$RC(i) = w(i)(\Sigma w)(i) / \sqrt{(w^{T}\Sigma w)} \quad (14)$$

Risk contributions; HRP follows its existing clustering allocation.

$$L1(t) = \Sigma_i |w(i,t) - w-(i,t)| ; one-way\ turnover(t) = L1(t)/2 \quad (15)$$

The two turnover labels must not be interchanged.

$$Cost(t) = L1(t) \times cside(t) = turnover(t) \times 2cside(t) \quad (16)$$

Corrects ambiguous historical half-L1 cost notation; preserves actual fee implementation.

$$Rnet(t) = Rgross(t) - Cost(t) ; V(t) = V(t-1)[1 + Rnet(t)] \quad (17)$$

Existing NAV deduction convention, initial V(0)=1.

$$VaR\alpha(L) = inf\{\ell : P(L \le \ell) \ge \alpha\} \quad (18)$$

Reported historical return quantile is converted to positive loss magnitude.

$$TailMean95 = -mean\{R(t) : R(t) \le q0.05(R)\} \quad (19)$$

Reported inclusive empirical tail mean; fractional-tail CVaR optimization is a distinct objective.

$$\sigma annual = sd(Rcycle)\sqrt{K} \quad (20)$$

K=12 for monthly cycle reporting; daily factor conventions differ.

$$Sharpe = mean(Rcycle - Rf,cycle) / sd(Rcycle - Rf,cycle) \times \sqrt{12} \quad (21)$$

Cash is compounded on the identical (entry,exit] interval.

$$MDD = min_t [V(t) / max_0 \le s \le t\ V(s) - 1] ; V(0) = 1 \quad (22)$$

Initial NAV is included in every running peak.

$$Y(i,t) = \prod_{s=1}^{21}[1 + r(i,t+s)] - \prod_{s=1}^{21}[1 + r(SPY,t+s)] \quad (23)$$

The same complete common-session horizon is required.

$$Ridge: min\beta\ ||y - X\beta||^2_2 + \alpha||\beta||^2_2 \quad (24)$$

Unpenalized intercept and training-only scaling follow the existing estimator.

$$ElasticNet: min\beta\ ||y - X\beta||^2_2/(2n) + \alpha a||\beta||_1 + \alpha(1-a)||\beta||^2_2/2 \quad (25)$$

Clarifies the actual estimator normalization; historical formula omitted these factors. No estimator changes.

$$IC(t) = Spearman(\hat{y}(i,t), Y(i,t)) ; meanIC = mean_t\, IC(t) \quad (26)$$

Constant vectors are undefined; bootstrap targets mean date IC, not pooled cross-date ranks.

The optimizer CVaR objective minimizes $z + \Sigma_j \max(L_j - z, 0)/[(1-\alpha)N]$ over its existing scenarios and constraints, with $\alpha=0.95$. It is distinct from the two-observation inclusive historical tail mean in Table XI. This clarification changes no objective or financial setting.

Appendix L Numerical provenance statement

PHASE4_CLAIM_TRACEABILITY.csv maps retained and revised claims to units, samples, publications and evidence, distinguishing empirical values, arithmetic, constants, structural numbering and bibliographic metadata. Historical ledger labels are not independent verification; omitted legacy cells remain traceable. PHASE4_PUBLICATION_SOURCE_MANIFEST.json binds manuscript hashes, historical sources, parent identities and revision evidence. This new editable DOCX was reconstructed from the supplied SSRN PDF; related older DOCX sources were inspected. Historical manuscripts and publications remain unchanged. Source limitations and author-confirmation requirements remain disclosed in the review report. The containing evidence commit identifies this revision without a self-referential hash.